generative AI *(continued)*
job loss and 16
limitations of 15
real-world uses of 5
reliability of outputs 16–17
generative AI (artificial intelligence) web applications
assessing first iteration 42
building major components 28–42
designing for better user experience 26
example of 20
generative AI lifecycle 26
project goal and requirements 21–26
generative AI web application, Next.js 43
configuration on 45–52
routing and configuration on 45–52
routing on 45–52
running project 44
setting up 44
generative chatbots for automated customer support 5
Google AI APIs 356
grounding, adding support 225–230
types of grounding 225
with log probabilities 227–230

H

hooks, defined 28
HTTP 429 (Too Many Requests) error 247
Hugging Face Spaces, alternative deployments 300
HumanMessage 188, 348
HyDE (hypothetical document embedding) 218

I

image
component 52
comprehension 80
generation 4, 5
processing 80
incremental delivery 63
incremental static regeneration 52
individual blog posts 48
information retrieval 180
input validation 269–275
Next.js API handler 271
overview 270–273
server vs. client validation 271
threat model 270
insufficient_quota error 26
integrating web apps with MCP
architecture overview 335
benefits of using MCP for web applications with LLMs 340
building end-to-end integration with MCP in Next.js 336–340
connecting Next.js and Vercel AI SDK with 334–341
integration testing, in React and Next.js 256–257
InterviewSidebar component 312
iterative refinement (reprompting) 115

J

JSX (JavaScript XML) 28

K

knowledge base grounding 225
knowledgebase routes 324
Kubernetes 291

L

LangChain agent 346
LangChain.js 269
agents 188–194
building AI workflows with 166–178, 194
integrating MCP servers with 345–349
memory components 185–188
modules 194
testing 264–267
troubleshooting 250–256
LangGraph vs. LCEL 195
language model specification 74
LanguageModelV1 interface 249
layout components 48
lazy loading 52
LCEL (LangChain Expression Language) 168, 195
Live debuggers 237
LLMLingua 212
LLMs (large language models) 3, 7, 19, 130, 166, 268, 332
techniques 164
log probabilities, grounding with 226–230
LTS (long-term support) 23